CLEAR: Cross-Modal Redundancy Control for Multimodal Recommendation

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Abstract

Multimodal recommendation improves collaborative filtering by incorporating heterogeneous item content. Existing methods commonly enhance multimodal fusion through stronger cross-modal interaction or alignment. However, visual and textual features often encode substantially overlapping item information, resulting in cross-modal redundancy. During representation learning and fusion, these repeatedly shared signals can dominate item representations, weakening the contribution of modality-specific information and limiting the benefits of multimodal modeling. To address this problem, we propose CLEAR, a lightweight and plug-and-play framework for cross-modal redundancy control in multimodal recommendation. CLEAR constructs the mean-centered cross-modal covariance between visual and textual representations and applies singular value decomposition to identify candidate directions where repeated cross-modal variation is most strongly concentrated. It then performs soft null-space projection to controllably attenuate these redundancy-dominant directions, while preserving their orthogonal components and retaining the selected shared information at a controlled scale. By directly reshaping the cross-modal representation geometry, CLEAR prevents repeatedly encoded information from dominating multimodal fusion without indiscriminately removing useful shared semantics. Moreover, CLEAR requires neither architectural modifications nor auxiliary training objectives, allowing it to be seamlessly integrated into existing multimodal recommenders. Extensive experiments on four public benchmark datasets and three representative backbones demonstrate that CLEAR consistently improves recommendation performance. Further analyses show that CLEAR effectively attenuates dominant cross-modal redundancy with modest computational overhead.

Introduction

Multimodal recommendation has emerged as a widely adopted paradigm for enhancing collaborative filtering by incorporating auxiliary information from multiple modalities (He and McAuley 2016; Liu et al. 2024). By leveraging heterogeneous item content beyond user-item interactions, multimodal recommenders aim to learn richer and more informative item representations. Existing approaches typically exploit multimodal content in two complementary ways: enriching item representations with semantic features extracted from individual modalities (He and McAuley 2016; Chen et al. 2017), and modeling latent item-item structures induced by modality-specific similarities (Zhang et al. 2021). Recent studies further introduce cross-modal interaction or alignment objectives to encourage consistent representations across modalities (Zhang et al. 2022; Liu et al. 2023). Despite their architectural differences, these methods largely share the premise that multimodal recommendation can be improved by extracting, propagating, or aligning information more effectively across modalities.

However, stronger cross-modal interaction does not necessarily lead to more effective information utilization. Visual and textual features often describe overlapping item attributes and therefore encode substantially repeated information (Zhang et al. 2023; Wang et al. 2025). Such shared information can be useful for recommendation; the problem arises when it is repeatedly aggregated across modalities and receives disproportionate influence during representation learning. In particular, when repeated cross-modal variation is concentrated in a few dominant directions, these directions can be continuously reinforced during multimodal fusion, weakening the relative contribution of less dominant but complementary modality-specific information. Alignment-driven objectives, while effective in establishing cross-modal correspondence, may further amplify already dominant shared components (Chaudhuri et al. 2025; Yi, Douady, and Chen 2025). Accordingly, multimodal recommendation faces a complementary challenge beyond strengthening cross-modal interaction: preventing repeatedly encoded information from becoming disproportionately dominant during multimodal representation learning.

Crucially, our empirical analysis reveals that substantial cross-modal redundancy is already present in pretrained visual and textual features, before any joint multimodal modeling takes place. As illustrated in Figure 1, given an anchor item, we retrieve neighboring items separately using the two modalities. The resulting similarity-score distributions exhibit substantial overlap, while the corresponding Top-K item neighborhoods overlap consistently more than expected under random retrieval. These results show that pretrained visual and textual features already encode highly repetitive item–item relational structures. Such structural overlap does not imply that all shared information is irrelevant. Instead, it means that downstream multimodal recommenders repeatedly receive correlated relational signals from both modali-

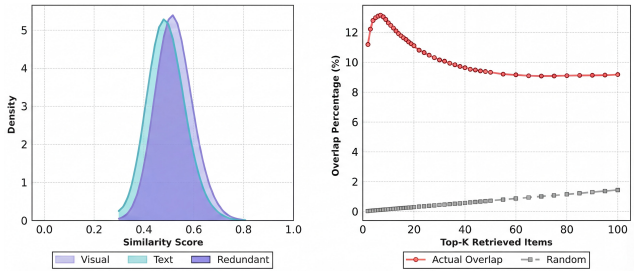


Figure 1: Empirical evidence of repeated cross-modal structure in pretrained item features on baby dataset. Left: density distributions of visual- and textual-similarity scores. Right: overlap ratios between the Top-K item neighborhoods induced by the two modalities.

ties, increasing the risk that already prominent shared structure dominates less represented modality-specific information. Together with the performance saturation observed when additional modalities are introduced into already informative recommenders (Yang et al. 2025; Ye et al. 2025), these findings motivate a different principle for multimodal fusion: controlling the relative dominance of repeated cross-modal information while retaining useful shared semantics.

Motivated by this perspective, we propose **C**ross-moda**L** r**E**dundancy **A**ttenuation via soft null-space p**R**ojection (CLEAR), a lightweight and plug-and-play framework for cross-modal redundancy control in multimodal recommendation. CLEAR first constructs the mean-centered cross-modal covariance between visual and textual representations to characterize their shared item-wise variation. It then applies singular value decomposition to identify the dominant visual and textual directions along which cross-modal variation is most strongly concentrated. These directions form a compact candidate subspace that captures the most prominent repeated structure across modalities.

Since dominant cross-modal variation may still contain useful information, CLEAR does not remove the identified subspace entirely. Instead, it applies soft null-space projection to controllably attenuate its contribution while leaving the orthogonal components unchanged. In this way, CLEAR reduces the dominance of repeatedly encoded cross-modal information without imposing a rigid shared-specific decomposition or indiscriminately discarding useful shared semantics. Operating directly at the representation level, CLEAR requires neither architectural modifications nor auxiliary training objectives and can be seamlessly integrated into existing multimodal recommenders. Experiments on four public benchmarks and three representative backbones demonstrate its effectiveness, efficiency, and broad applicability. Our contributions are summarized as follows:

- We reveal cross-modal redundancy dominance as an underexplored limitation of multimodal recommendation. Beyond insufficient interaction or alignment, repeatedly encoded cross-modal information can receive disproportionate influence during multimodal fusion, weakening the utilization of complementary modality-specific information.
- We propose CLEAR, a lightweight geometric solution principle that models repeated cross-modal variation through mean-centered cross-modal covariance and identifies redundancy-dominant directions via singular value decomposition. Paired soft null-space projection then controllably attenuates these directions while preserving their orthogonal components and retaining useful shared information.
- CLEAR operates directly at the representation level and requires neither architectural modifications nor auxiliary objectives. Experiments on four public benchmarks, complemented by integration into three representative multimodal recommenders, demonstrate its effectiveness, architectural compatibility, and computational efficiency.

Related Work

Model Architectures in MMRec

Model-level methods focus on designing architectures to capture interactions among users, items, and multimodal features (Pan et al. 2025). Early studies primarily employ Graph Convolutional Networks (GCNs) to propagate modality-aware signals. MMGCN (Wei et al. 2019) constructs modality-specific user–item graphs, while LATTICE (Zhang et al. 2021) learns latent item–item structures from multimodal similarities. Subsequent methods improve the robustness or efficiency of this paradigm. FREEDOM (Zhou and Shen 2023) suppresses noise in multimodal graphs through degree-sensitive edge pruning, whereas TMLP (Huang et al. 2025) replaces expensive graph propagation with lightweight feature transformations. Although these methods improve downstream propagation and fusion, they do not directly regulate repeatedly encoded cross-modal variations in the input representation space. In contrast to these architecture-specific designs, CLEAR is a lightweight representation-level framework that can be integrated into different multimodal recommenders without redesigning their propagation or fusion architectures.

Optimization Strategies in MMRec

Optimization-level methods shape multimodal representations through self-supervised, alignment, denoising, or disentanglement objectives. SLMRec (Tao et al. 2022) and BM3 (Zhou et al. 2023b) introduce self-supervised objectives to enhance the interaction between modality representations and collaborative signals. Other studies reconsider the role of ID embeddings or employ alternative alignment objectives to alleviate ID bias and improve cross-modal convergence (Yuan et al. 2023; Li et al. 2025). Beyond alignment, several methods attempt to reduce modality interference through feature purification, decorrelation, or disentanglement. MGNM (Mo et al. 2024) combines a dynamic de-redundancy loss that penalizes feature correlations induced by stacked GNN layers with modality-guided feature purifiers for suppressing preference-irrelevant modality noise. MODEST (Zhang et al. 2023) reduces potentially spurious cross-modal dependence through adaptive sample reweighting and modality decorrelation. DHMRec (Zhan et al. 2026) explicitly separates modality-common and modality-specific fea-

tures before hierarchical fusion. These methods are related to CLEAR in their goal of reducing cross-modal interference, but generally rely on sample reweighting, auxiliary optimization objectives, or dedicated disentanglement architectures. In contrast, CLEAR estimates redundancy-dominant candidate directions directly from mean-centered cross-modal covariance and regulates their influence through a soft geometric projection, without predefining shared–specific semantic boundaries or modifying the original recommendation objective.

Null-Space Projection in Machine Learning

Null-space projection regulates selected representation directions by operating on their orthogonal complement and has been applied to fairness learning, knowledge editing, and recommendation. INLP (Ravfogel et al. 2020) projects representations onto the null space of classifiers that encode sensitive attributes, thereby reducing sensitive correlations. AlphaEdit (Fang et al. 2024) constrains parameter updates to the null space of preserved knowledge representations during knowledge editing. In recommendation, AlphaFuse (Hu et al. 2025) injects collaborative signals into the null space of language embeddings to incorporate behavioral information while limiting interference with the original semantic space. Adam-NSCL (Wang et al. 2021b) projects candidate parameter updates onto layer-wise approximate null spaces derived from the feature covariance of previous tasks, thereby mitigating catastrophic forgetting in continual learning.

Preliminaries

Problem Definition

In multimodal recommendation, the goal is to improve preference prediction by jointly exploiting user–item interactions and item content. Let $\mathcal{U} = \{u_1, \dots, u_M\}$ and $\mathcal{I} = \{i_1, \dots, i_N\}$ denote the user and item sets. The observed implicit feedback is represented by $\mathbf{R} \in \{0,1\}^{M \times N}$, where $r_{ui} = 1$ indicates that user u has interacted with item i. Each item is associated with raw visual and textual features, collected as $\mathbf{V}^{\mathrm{raw}} \in \mathbb{R}^{N \times d_v^{\mathrm{raw}}}$ and $\mathbf{T}^{\mathrm{raw}} \in \mathbb{R}^{N \times d_t^{\mathrm{raw}}}$, respectively. The task is to learn user and item representations $\mathbf{e}_u$ and $\mathbf{e}_i$ such that a scoring function $s_\Theta(\mathbf{e}_u, \mathbf{e}_i)$ accurately predicts the preference score $\hat{y}_{ui}$.

Multimodal Recommendation Formulation

Multimodal recommenders commonly transform raw content features through modality-specific encoders:

$$\mathbf{V} = \phi_v(\mathbf{V}^{\mathrm{raw}}) \in \mathbb{R}^{N \times d_v}, \qquad \mathbf{T} = \phi_t(\mathbf{T}^{\mathrm{raw}}) \in \mathbb{R}^{N \times d_t}. \tag{1}$$

An existing recommendation backbone g_Θ then combines the encoded modality features with the interaction structure to produce user and item representations:

$$(\mathbf{E}_\mathcal{U}, \mathbf{E}_\mathcal{I}) = g_\Theta(\mathbf{R}, \mathbf{V}, \mathbf{T}), \qquad \hat{y}_{ui} = s_\Theta(\mathbf{e}_u, \mathbf{e}_i). \tag{2}$$

Depending on the backbone, g_Θ may contain graph propagation, attention, self-supervised learning, or other modality-fusion operations. CLEAR is inserted between the modality encoders and this original backbone, regulating cross-modal representation structure without changing the backbone-specific scoring function or training objective.

Methodology

In this section, we present CLEAR, a geometric module that regulates cross-modal redundancy through controlled soft null-space projection. Figure 2 illustrates the overall workflow.

Overview of CLEAR

CLEAR addresses cross-modal redundancy by performing a direct geometric intervention on the learned representations rather than relying on auxiliary losses. The method begins by explicitly modeling the co-variation between visual and textual features through their cross-modal covariance matrix. Then we perform SVD on this covariance matrix to identify the dominant directions in which redundant information is concentrated. CLEAR applies a soft null-space projection that attenuates components along those directions while fully preserving information in the orthogonal complement so that modality-specific signals become more prominent and less overshadowed by shared semantics. The projected features are then seamlessly fed into the downstream graph neural network to capture collaborative filtering signals. In the following section, we will provide a detailed explanation of CLEAR.

Cross-Modal Redundancy Modeling

We use *cross-modal redundancy* to describe repeatedly encoded cross-modal variation whose disproportionate accumulation can reduce the relative utilization of modality-specific information during fusion. This definition concerns the allocation of representational influence rather than the mere existence of shared semantics. Accordingly, strong cross-modal co-variation is treated as evidence for a candidate redundancy-dominant subspace, not as proof that the corresponding information is useless for recommendation.

Mean-Centered Cross-Modal Covariance. Given transformed visual and textual feature matrices $\mathbf{V} \in \mathbb{R}^{N \times d_v}$ and $\mathbf{T} \in \mathbb{R}^{N \times d_t}$, we define their modality-wise means and centered representations as

$$\begin{aligned} \boldsymbol{\mu}_v &= \frac{1}{N}\mathbf{V}^\top \mathbf{1}_N, \quad \bar{\mathbf{V}} = \mathbf{V} - \mathbf{1}_N \boldsymbol{\mu}_v^\top, \\ \boldsymbol{\mu}_t &= \frac{1}{N}\mathbf{T}^\top \mathbf{1}_N, \quad \bar{\mathbf{T}} = \mathbf{T} - \mathbf{1}_N \boldsymbol{\mu}_t^\top. \end{aligned} \tag{3}$$

Mean-centering removes modality-wise first-order offsets, allowing the subsequent estimator to capture shared item-wise variation rather than global feature bias. The centered cross-modal covariance is

$$\mathbf{C} = \frac{1}{N}\bar{\mathbf{V}}^\top \bar{\mathbf{T}} \in \mathbb{R}^{d_v \times d_t}. \tag{4}$$

Its element c_{ij} measures how the i-th visual dimension and the j-th textual dimension vary together across the N items.

Candidate Directions via SVD. Let $r = \min(d_v, d_t)$. We use the thin SVD

$$\mathbf{C} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{W}^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{w}_i^\top, \tag{5}$$

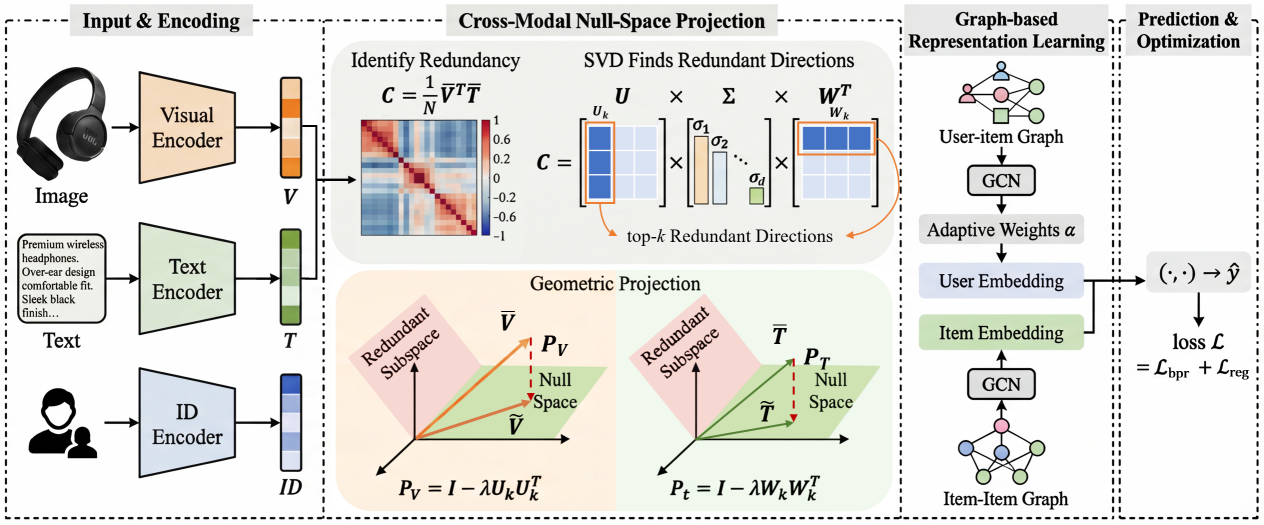


Figure 2: The overall architecture of CLEAR. Visual and textual features are encoded, and a cross-modal covariance matrix $\mathbf{C}$ is constructed and decomposed via SVD to identify the top-k redundancy-dominant candidate directions. Soft null-space projections $\mathbf{P}_v$ and $\mathbf{P}_t$ attenuate the influence of these directions while preserving modality-specific components for the illustrated downstream graph-based recommender.

where $\mathbf{U} \in \mathbb{R}^{d_v \times r}$, $\mathbf{W} \in \mathbb{R}^{d_t \times r}$, and $\sigma_1 \geq \cdots \geq \sigma_r \geq 0$. Repeated shared variation affects many items coherently and therefore tends to concentrate covariance energy in a small number of leading paired directions, which we refer to as the *low-rank redundancy concentration* hypothesis. The long-tailed spectra in Figure 5 provide empirical support for this structure. A larger singular value indicates stronger paired cross-modal co-variation, but does not by itself imply that the associated information is task-irrelevant. We therefore regard $\mathbf{U}_k = [\mathbf{u}_1, \ldots, \mathbf{u}_k]$ and $\mathbf{W}_k = [\mathbf{w}_1, \ldots, \mathbf{w}_k]$ as paired candidate directions whose influence should be regulated rather than indiscriminately removed.

Controlled Soft Null-Space Projection

The hard null-space projection is recovered when all components in the selected candidate subspaces are removed. CLEAR instead uses the following soft projectors:

$$\begin{aligned} \mathbf{P}_v &= \mathbf{I}_{d_v} - \lambda \mathbf{U}_k \mathbf{U}_k^\top, \\ \mathbf{P}_t &= \mathbf{I}_{d_t} - \lambda \mathbf{W}_k \mathbf{W}_k^\top, \end{aligned} \qquad 0 \leq \lambda \leq 1. \tag{6}$$

The projected features are

$$\widetilde{\mathbf{V}} = \bar{\mathbf{V}} \mathbf{P}_v, \qquad \widetilde{\mathbf{T}} = \bar{\mathbf{T}} \mathbf{P}_t. \tag{7}$$

When $\lambda = 0$, the projection leaves the centered features unchanged; when $\lambda = 1$, it becomes a hard null-space projection. For $0 < \lambda < 1$, selected components are attenuated rather than erased.

Low-Rank Structure of Cross-Modal Redundancy The redundancy-dominant spectrum is primarily governed by the dataset-level cross-modal feature distribution, so the effective scope of redundancy control is naturally data-dependent. A dynamic energy-ratio rule can change the selected rank whenever the singular-value distribution fluctuates during training, thereby changing the projection target and introducing avoidable optimization instability. We therefore treat k as a dataset-specific control hyperparameter: it is selected on the validation set and then fixed during training, while λ controls the attenuation strength. This design assumes that the dominant redundancy structure is sufficiently stable to support a consistent control scope, rather than requiring the exact mathematical rank of redundancy to be immutable. The comparison with dynamic ratio selection in ablation study empirically supports this choice.

Theoretical Analysis

We next establish why repeated shared components can become disproportionately influential and what the soft projection changes. The analysis characterizes representation geometry; it does not assume that a dominant shared component is semantically useless.

Proposition 1 (Amplification of a Repeated Shared Component). Suppose two representations in a common fusion space admit $\mathbf{v} = \mathbf{s} + \mathbf{a}$ and $\mathbf{t} = \mathbf{s} + \mathbf{b}$, where the modality-specific components $\mathbf{a}$ and $\mathbf{b}$ are uncorrelated with the repeated shared component $\mathbf{s}$ and with each other. For non-negative fusion weights α and β,

$$\begin{aligned} \mathbb{E}\|\alpha\mathbf{v} + \beta\mathbf{t}\|_2^2 = (\alpha+\beta)^2 \mathbb{E}\|\mathbf{s}\|_2^2 \\ + \alpha^2 \mathbb{E}\|\mathbf{a}\|_2^2 + \beta^2 \mathbb{E}\|\mathbf{b}\|_2^2. \end{aligned} \tag{8}$$

Thus, fusion assigns the repeated component an additional contribution $2\alpha\beta\mathbb{E}\|\mathbf{s}\|_2^2$. This result identifies an amplification mechanism without labeling the shared component as task-irrelevant.

Proposition 2 (Controlled Component Retention). For a visual row vector $\mathbf{x}$, let $\mathbf{x}_\parallel = \mathbf{x}\mathbf{U}_k\mathbf{U}_k^\top$ and $\mathbf{x}_\perp = \mathbf{x}(\mathbf{I}_{d_v} -$

Table 1: Statistics of the experimental datasets.

Dataset	#Users	#Items	#Inters	Sparsity
Baby	19,445	7,050	160,792	99.88%
Sports	35,598	18,357	296,337	99.95%
Clothing	39,387	23,033	278,677	99.97%
MicroLens	98,129	17,228	705,174	99.96%

$\mathbf{U}_k\mathbf{U}_k^\top$). Then

$$\mathbf{x}\mathbf{P}_v = (1-\lambda)\mathbf{x}_\parallel + \mathbf{x}_\perp. \tag{9}$$

The orthogonal component is preserved exactly, while the selected component is retained with scale $1-\lambda$. The textual projector satisfies the same property.

Proposition 3 (Exact Paired-Direction Attenuation). Let $\widetilde{\mathbf{C}} = \frac{1}{N}\widetilde{\mathbf{V}}^\top\widetilde{\mathbf{T}}$. Applying the paired soft projectors in Eq. (6) gives

$$\begin{aligned}\widetilde{\mathbf{C}} = &\sum_{i=1}^{k}(1-\lambda)^2\sigma_i\mathbf{u}_i\mathbf{w}_i^\top \\ &+ \sum_{i=k+1}^{r}\sigma_i\mathbf{u}_i\mathbf{w}_i^\top.\end{aligned} \tag{10}$$

Consequently, CLEAR scales only the selected paired directions and leaves all remaining singular directions unchanged. Complete proofs are provided in the appendix.

Plug-and-Play Integration and Training

Let g_Θ denote an arbitrary multimodal recommendation backbone. CLEAR replaces its original modality inputs with the projected features at the same interface:

$$\begin{aligned}(\mathbf{E}_\mathcal{U}, \mathbf{E}_\mathcal{I}) &= g_\Theta(\mathbf{R}, \widetilde{\mathbf{V}}, \widetilde{\mathbf{T}}), \\ \widehat{y}_{ui} &= s_\Theta(\mathbf{e}_u, \mathbf{e}_i).\end{aligned} \tag{11}$$

In particular, CLEAR preserves the host model's original objective $\mathcal{L}_{\text{base}}$ and introduces no additional loss term.This design allows the model to learn representations that are both discriminative for recommendation and compatible with redundancy suppression.

At the start of training and every τ epochs thereafter, we recompute the global feature means and paired singular directions from the current encoded features. The covariance used to construct the projection basis is detached from the computation graph, and the resulting quantities are cached between updates. During ordinary mini-batch optimization, gradients continue to flow through the encoded and projected features, while the cached projection basis is treated as fixed. This procedure allows CLEAR to be integrated into different backbones without changing their architectures or training objectives. Training prcedure and pseudocode are shown in Appendix.

Experiments

Experimental Settings

We evaluate CLEAR against 14 baselines on four public benchmark datasets using two ranking metrics.

Datasets. We use three widely adopted Amazon review datasets (McAuley et al. 2015), namely Baby, Sports, and Clothing, together with the large scale MicroLens micro-video dataset. Table 1 summarizes all four datasets. Dataset-specific preprocessing details are provided in the appendix.

Baselines. We compare with three general collaborative filtering methods: MF-BPR (Rendle et al. 2012), LightGCN (He et al. 2020), and LayerGCN (Zhou et al. 2023a). The eleven multimodal baselines include VBPR (He and McAuley 2016), MMGCN (Wei et al. 2019), DualGNN (Wang et al. 2021a), LATTICE (Zhang et al. 2021), FREEDOM (Zhou and Shen 2023), MGCN (Yu et al. 2023), SLMRec (Tao et al. 2022), MICRO (Zhang et al. 2022), BM3 (Zhou et al. 2023b), LGMRec (Guo et al. 2024), and MENTOR (Xu et al. 2025). Further implementation and baseline details are provided in the appendix.

Evaluation Protocol. We evaluate ranking quality using Recall@K and NDCG@K for $K \in \{10, 20\}$. Interactions are split into training, validation, and test sets with an 8:1:1 ratio. Hyperparameters are selected on the validation set, and final performance is reported on the test set.

Implementation Details. We implement CLEAR with the open-source MMRec framework (Zhou 2023) and follow the standard preprocessing and ranking protocol. Our runs are implemented in PyTorch on a single NVIDIA RTX 4090 GPU with 24 GB of memory. We use 64-dimensional user and item embeddings, Xavier initialization (Glorot and Bengio 2010), and Adam (Kingma and Ba 2014). For CLEAR, we search $\lambda \in \{0.3, 0.5, 0.7, 0.8, 0.9\}$ and $k \in \{1, 2, 4, 8, 16, 20\}$; both are selected on the validation set, and k is then kept fixed during training.

Overall Performance

We conduct our CLEAR framework under MENTOR backbone. Table 2 compares CLEAR with the baselines on all four datasets. We draw three observations.

Different model families remain competitive across datasets. Alignment-based, self-supervised, graph-based, and disentanglement-based recommenders exhibit different relative rankings across datasets. This variation indicates that architectural complexity or stronger alignment alone does not determine how effectively pretrained multimodal features are used, leaving room for a representation-level mechanism that complements different model families.

Controlled redundancy attenuation is effective across model families. CLEAR neither enforces additional cross-modal alignment nor prespecifies a shared–specific decomposition. It estimates redundancy-dominant candidate directions from mean-centered cross-modal covariance and attenuates their relative influence while retaining them at a controlled scale. CLEAR outperforms representative alignment- and disentanglement-based methods on the three Amazon datasets, supporting that redundancy control is complementary to existing multimodal modeling strategies.

CLEAR achieves the strongest overall performance. On Recall@10, CLEAR improves over the strongest available baseline by 6.0%, 2.7%, 2.4%, and 2.8% on Baby,

Table 2: Performance comparison of baselines and CLEAR in terms of Recall@K (R@K) and NDCG@K (N@K). Superscript * denotes a statistically significant improvement over the best-performing baseline with $p < 0.05$, based on a paired t-test across repeated runs using matched random seeds.

Datasets	Baby				Sports				Clothing				MicroLens			
Methods	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**
MF-BPR	0.0357	0.0575	0.0192	0.0249	0.0432	0.0653	0.0241	0.0298	0.0187	0.0279	0.0103	0.0126	0.0624	0.0959	0.0322	0.0408
LightGCN	0.0479	0.0754	0.0257	0.0328	0.0569	0.0864	0.0311	0.0387	0.0340	0.0526	0.0188	0.0236	0.0720	0.1075	0.0376	0.0467
LayerGCN	0.0529	0.0820	0.0281	0.0355	0.0594	0.0916	0.0323	0.0406	0.0371	0.0566	0.0200	0.0247	0.0735	0.1137	0.0376	0.0479
VBPR	0.0423	0.0663	0.0223	0.0284	0.0558	0.0856	0.0307	0.0384	0.0281	0.0415	0.0158	0.0192	0.0677	0.1026	0.0351	0.0441
MMGCN	0.0378	0.0615	0.0200	0.0261	0.0370	0.0605	0.0193	0.0254	0.0218	0.0345	0.0110	0.0142	0.0421	0.0701	0.0207	0.0279
DualGNN	0.0448	0.0716	0.0240	0.0309	0.0568	0.0859	0.0310	0.0385	0.0454	0.0683	0.0241	0.0299	0.0763	0.1132	0.0394	0.0489
LATTICE	0.0547	0.0850	0.0292	0.0370	0.0620	0.0953	0.0335	0.0421	0.0492	0.0733	0.0268	0.0330	0.0726	0.1089	0.0380	0.0473
SLMRec	0.0529	0.0775	0.0290	0.0353	0.0663	0.0990	0.0365	0.0450	0.0452	0.0675	0.0247	0.0303	0.0778	0.1190	0.0405	0.0511
MICRO	0.0570	0.0905	0.0310	0.0406	0.0675	0.1026	0.0365	0.0463	0.0496	0.0743	0.0264	0.0332	0.0706	0.1101	0.0360	0.0461
BM3	0.0564	0.0883	0.0301	0.0383	0.0656	0.0980	0.0355	0.0438	0.0422	0.0621	0.0231	0.0281	0.0606	0.0981	0.0304	0.0400
FREEDOM	0.0627	0.0992	0.0330	0.0424	0.0717	0.1089	0.0385	0.0481	0.0629	0.0941	0.0341	0.0420	0.0674	0.1032	0.0345	0.0437
MGCN	0.0620	0.0964	0.0339	0.0427	0.0729	0.1106	0.0397	0.0496	0.0641	0.0945	0.0347	0.0428	0.0759	0.1148	0.0387	0.0486
LGMRec	0.0644	0.1002	0.0349	0.0440	0.0720	0.1068	0.0390	0.0480	0.0555	0.0828	0.0302	0.0371	0.0748	0.1132	0.0390	0.0489
MENTOR	0.0648	0.1035	0.0349	0.0448	0.0736	0.1130	0.0399	0.0501	0.0659	0.0976	0.0357	0.0437	0.0785	0.1164	0.0410	0.0508
CLEAR	**0.0687***	**0.1045***	**0.0364***	**0.0456***	**0.0756***	**0.1146***	**0.0411***	**0.0511***	**0.0675***	**0.1001***	**0.0358***	**0.0440***	**0.0807***	**0.1194***	**0.0419***	**0.0518***

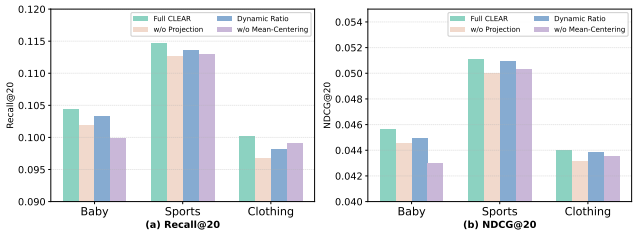


Figure 3: Ablation study comparing four variants on three datasets. The bars report R@20 and N@20 for Full CLEAR, w/o projection, Dynamic Ratio, and w/o mean-centering.

Sports, Clothing, and MicroLens, respectively, with consistent gains on the remaining metrics. To further examine whether the benefit of CLEAR is associated with the degree of shared cross-modal structure in the input features, we use linear Centered Kernel Alignment (CKA) as a representation-level measure of structural similarity between pretrained visual and textual features. On Baby, Sports, and MicroLens, the measured CKA values are 0.26, 0.20, and 0.12, while the corresponding Recall@10 gains over the same unprojected backbone are 4.25%, 2.16%, and 1.51%, respectively. The gain decreases together with CKA across these datasets, which is consistent with CLEAR providing larger benefits when the two modalities contain stronger shared structure. The formal definition of CKA, complete results, and further discussion are provided in the appendix.

Ablation Study

We compare four configurations on Baby, Sports, and Clothing: (1) full CLEAR with mean-centered covariance and fixed top-k selection, (2) removing null-space projection, (3) replacing fixed k with a dynamic singular-value-energy ratio, and (4) removing mean-centering.

Projection and rank selection. Removing the projection consistently reduces R@10 and N@10 on all three datasets, supporting the role of regulating dominant cross-modal covariation. This result does not imply that the selected directions are task-irrelevant: the soft projection retains them with scale $1 - \lambda$ and leaves their orthogonal components unchanged. Fixed k also consistently outperforms dynamic ratio selection in our evaluation. A fixed rank provides a stable control target while the singular-value spectrum evolves, without requiring the exact redundancy rank to be an immutable property of the dataset.

Table 3: Comparison of projection direction and strategies. Top-k SVD corresponds to CLEAR; k, λ are using the optimal setting to ensure a fair outcome.

	Baby		Sports		Clothing	
Strategy	**R@20**	**N@20**	**R@20**	**N@20**	**R@20**	**N@20**
Top-k SVD	**0.1045**	**0.0456**	**0.1146**	**0.0511**	**0.1001**	**0.0440**
Random-k	0.1023	0.0446	0.1107	0.0491	0.0955	0.0428
Bottom-k	0.1015	0.0444	0.1097	0.0492	0.0955	0.0426
CCA	0.1011	0.0441	0.1117	0.0502	0.0967	0.0432
PLS	0.1031	0.0446	0.1140	**0.0511**	0.0978	0.0437

Mean-centering. Removing mean-centering decreases R@10 by 3.78%, 0.66%, and 1.33% and N@10 by 2.20%, 0.97%, and 1.12% on Baby, Sports, and Clothing, respectively. Mean-centering removes modality-wise first-order offsets before covariance estimation, helping the selected directions reflect shared item-wise variation rather than global feature bias.

Projection direction construction and method selection. Replacing top-k directions with random-k or bottom-k directions consistently degrades performance, confirming that the location of the attenuated subspace is essential. We further compare CLEAR with Canonical Correlation Analysis (CCA), which identifies directions with maximum normalized cross-modal correlation, and Partial Least Squares (PLS), which extracts latent directions by maximizing cross-modal covariance. Both alternatives underperform the paired covariance SVD used by CLEAR, indicating that its simultaneous extraction of covariance-dominant directions is better aligned with our redundancy-control objective.

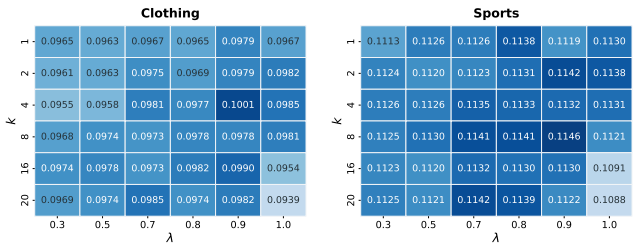


Figure 4: Hyperparameter analysis on Clothing and Sports. Heatmaps show R@20 for different λ and k combinations; darker colors indicate better performance.

Table 4: Generalization analysis on the Baby dataset.

	Base		+ CLEAR		Gain (%)	
Backbone	**R@10**	**N@10**	**R@10**	**N@10**	**R@10**	**N@10**
VBPR	0.0423	0.0223	**0.0469**	**0.0258**	10.87	15.70
FREEDOM	0.0624	0.0328	**0.0634**	**0.0337**	1.60	2.74
MENTOR	0.0648	0.0349	**0.0687**	**0.0364**	6.02	4.30

Hyperparameter Analysis

Projection strength λ. Performance generally improves as λ increases within the evaluated range and is best at $\lambda = 0.9$ on both datasets. In contrast, the hard projection at $\lambda = 1$ completely removes the selected components and causes a marked performance decline.

Candidate-subspace rank k. Very small k regulates only a few dominant directions, whereas an excessively large k attenuates a broader portion of the shared representation space. Clothing performs best at $k = 4$, while Sports favors $k = 8$. CLEAR remains reasonably stable over a small validation grid, although poorly matched settings may either insufficiently suppress repeated components or attenuate useful shared information. Both hyperparameters are selected once on the validation set and kept fixed during training.

Efficiency, Generalization, and Visualization

Computational Efficiency. We analyze the computational overhead introduced by CLEAR framework. Assuming feature dimension d, the time complexity is dominated by cross-modal covariance construction and feature projection, both requiring $\mathcal{O}(Nd^2)$, together with the SVD decomposition requiring $\mathcal{O}(d^3)$, where N denotes the total number of items. The additional space complexity is $\mathcal{O}(d^2)$ for storing the covariance and projection matrices. Since $d = 64$ in our implementation, the practical overhead remains modest. Detailed measurements in appendix show that CLEAR remains competitive in both training efficiency and memory usage.

Generalization Analysis. We integrate CLEAR into three multimodal recommenders with distinct architectures: VBPR, FREEDOM, MENTOR. As shown in Table 4, CLEAR improves every backbone on both metrics. Although the magnitude depends on the host model, the consistent direction of improvement supports that CLEAR is not tied to a particular encoder or graph architecture. We provide more detailed experimental results in the appendix.

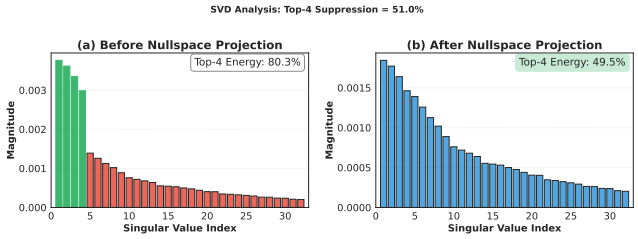


Figure 5: Singular-value distribution of the cross-modal covariance before and after soft null-space projection.

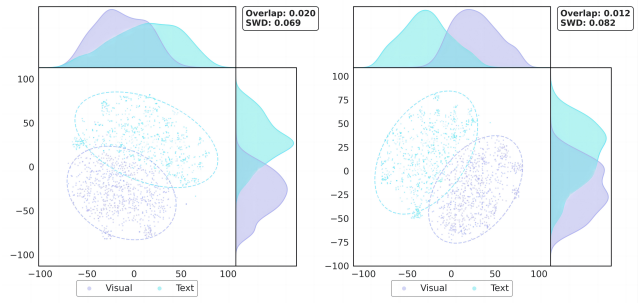


Figure 6: t-SNE visualization on Baby before and after soft null-space projection, together with overlap and Sliced Wasserstein Distance (SWD).

Visualization Analysis. Figure 5 shows that CLEAR attenuates the leading singular values and produces a flatter covariance spectrum, matching the exact paired-direction effect in Proposition 3. Figure 6 illustrates the distribution of projects before and after null-space projection. The top and right display the marginal distributions. Through overlap and SWD metrics, we can visually observe that our CLEAR method significantly reduces redundancy at the representation level. Theorem in appendix further proves this assertion

Conclusion

In this paper, we identify cross-modal redundancy as an underexplored problem in multimodal recommendation: repeatedly encoded shared variation can receive disproportionate influence during fusion and limit the relative utilization of modality-specific information. We propose CLEAR, a lightweight geometric module that estimates redundancy-dominant candidate directions from mean-centered cross-modal covariance and softly attenuates their influence through paired null-space projections. CLEAR does not predefine a shared–specific boundary, remove all shared information, or introduce an auxiliary loss; instead, it retains selected components at a controlled scale and preserves their orthogonal complements. Experiments on four datasets consistently support its effectiveness and plug-and-play applicability. Future work will investigate extending CLEAR beyond paired modalities by using generalized CCA or PLS to estimate a globally consistent shared subspace across visual, textual, temporal, and acoustic modalities, followed by modality-specific controlled null-space projections.

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Theoretical Analysis and Proofs

Proof of Shared-Component Amplification

We first prove Proposition 1. Substituting $\mathbf{v} = \mathbf{s} + \mathbf{a}$ and $\mathbf{t} = \mathbf{s} + \mathbf{b}$ into the positive weighted fusion gives

$$\begin{aligned}\mathbf{h} &= \alpha\mathbf{v} + \beta\mathbf{t} \\ &= (\alpha+\beta)\mathbf{s} + \alpha\mathbf{a} + \beta\mathbf{b}.\end{aligned} \tag{12}$$

Because $\mathbf{s}$, $\mathbf{a}$, and $\mathbf{b}$ are mutually uncorrelated, all cross-terms vanish. Therefore,

$$\begin{aligned}\mathbb{E}\|\mathbf{h}\|_2^2 = (\alpha+\beta)^2\mathbb{E}\|\mathbf{s}\|_2^2 \\ + \alpha^2\mathbb{E}\|\mathbf{a}\|_2^2 + \beta^2\mathbb{E}\|\mathbf{b}\|_2^2.\end{aligned} \tag{13}$$

If the two modality-wise shared contributions were counted separately, their total coefficient would be $\alpha^2 + \beta^2$. Fusion instead assigns the shared component the coefficient $(\alpha + \beta)^2$, producing the additional contribution $2\alpha\beta\mathbb{E}\|\mathbf{s}\|_2^2$. This proves the amplification mechanism without assuming that the shared component is task-irrelevant.

Proof of Controlled Component Retention

We next prove Proposition 2. Let $\mathbf{\Pi} = \mathbf{Q}_k\mathbf{Q}_k^\top$. Since $\mathbf{Q}_k$ has orthonormal columns, $\mathbf{\Pi}^\top = \mathbf{\Pi}$ and $\mathbf{\Pi}^2 = \mathbf{\Pi}$. Decompose $\mathbf{x}$ as

$$\begin{aligned}\mathbf{x}_\parallel &= \mathbf{x}\mathbf{\Pi}, \\ \mathbf{x}_\perp &= \mathbf{x}(\mathbf{I} - \mathbf{\Pi}), \\ \mathbf{x} &= \mathbf{x}_\parallel + \mathbf{x}_\perp.\end{aligned} \tag{14}$$

For $\mathbf{P} = \mathbf{I} - \lambda\mathbf{\Pi}$,

$$\begin{aligned}\mathbf{x}\mathbf{P} &= \mathbf{x} - \lambda\mathbf{x}\mathbf{\Pi} \\ &= (1-\lambda)\mathbf{x}_\parallel + \mathbf{x}_\perp.\end{aligned} \tag{15}$$

The resulting perturbation satisfies

$$\begin{aligned}\|\mathbf{x}\mathbf{P} - \mathbf{x}\|_2 &= \lambda\|\mathbf{x}_\parallel\|_2 \\ &\le \lambda\|\mathbf{x}\|_2.\end{aligned} \tag{16}$$

The inequality follows from the non-expansiveness of orthogonal projection. Thus the orthogonal component is preserved exactly, whereas the selected component is scaled by $1 - \lambda$.

Proof of Exact Cross-Modal Spectral Attenuation

We now prove Proposition 3. Because $\mathbf{P}_v$ and $\mathbf{P}_t$ are symmetric, the projected cross-modal covariance is

$$\begin{aligned}\widetilde{\mathbf{C}} &= \frac{1}{N}(\bar{\mathbf{V}}\mathbf{P}_v)^\top(\bar{\mathbf{T}}\mathbf{P}_t) \\ &= \mathbf{P}_v\mathbf{C}\mathbf{P}_t.\end{aligned} \tag{17}$$

For each singular-vector pair,

$$\begin{aligned}\mathbf{P}_v\mathbf{u}_i &= \begin{cases}(1-\lambda)\mathbf{u}_i, & i \le k, \\ \mathbf{u}_i, & i > k,\end{cases} \\ \mathbf{P}_t\mathbf{w}_i &= \begin{cases}(1-\lambda)\mathbf{w}_i, & i \le k, \\ \mathbf{w}_i, & i > k.\end{cases}\end{aligned} \tag{18}$$

Substituting $\mathbf{C} = \sum_{i=1}^r \sigma_i\mathbf{u}_i\mathbf{w}_i^\top$, where $r = \min(d_v, d_t)$, gives

$$\begin{aligned}\widetilde{\mathbf{C}} = \sum_{i=1}^{k}(1-\lambda)^2\sigma_i\mathbf{u}_i\mathbf{w}_i^\top \\ + \sum_{i=k+1}^{r}\sigma_i\mathbf{u}_i\mathbf{w}_i^\top.\end{aligned} \tag{19}$$

Because the rank-one matrices $\{\mathbf{u}_i\mathbf{w}_i^\top\}_{i=1}^r$ are orthonormal under the Frobenius inner product,

$$\begin{aligned}\|\widetilde{\mathbf{C}}\|_F^2 = (1-\lambda)^4\sum_{i=1}^{k}\sigma_i^2 \\ + \sum_{i=k+1}^{r}\sigma_i^2.\end{aligned} \tag{20}$$

Moreover,

$$\begin{aligned}\|\mathbf{C}\|_F^2 - \|\widetilde{\mathbf{C}}\|_F^2 = \left[1-(1-\lambda)^4\right] \\ \cdot\sum_{i=1}^{k}\sigma_i^2.\end{aligned} \tag{21}$$

This proves that CLEAR attenuates only the selected paired directions and leaves all remaining singular directions unchanged.

Preference Information and Recommendation Utility

Proposition 4 (Preference-Information Preservation). Let $Y_{uij} \in \{0, 1\}$ denote the pairwise preference target for user u and items i, j, and define

$$\mathbf{Z}_{ij} = \begin{bmatrix}\bar{\mathbf{v}}_i^\top & \bar{\mathbf{t}}_i^\top & \bar{\mathbf{v}}_j^\top & \bar{\mathbf{t}}_j^\top\end{bmatrix}^\top. \tag{22}$$

Let

$$\mathcal{P}_\lambda = \mathrm{blkdiag}(\mathbf{P}_v, \mathbf{P}_t, \mathbf{P}_v, \mathbf{P}_t), \qquad \widetilde{\mathbf{Z}}_{ij} = \mathcal{P}_\lambda\mathbf{Z}_{ij}. \tag{23}$$

Conditional on the cached projectors, for every $0 \le \lambda < 1$,

$$I(Y_{uij}; \widetilde{\mathbf{Z}}_{ij} \mid \mathcal{P}_\lambda) = I(Y_{uij}; \mathbf{Z}_{ij} \mid \mathcal{P}_\lambda). \tag{24}$$

For the hard projection $\lambda = 1$, write $\mathbf{Z}_{ij} = (\mathbf{Z}_{ij,\perp}, \mathbf{Z}_{ij,\parallel})$. Its exact preference-information loss is

$$\begin{aligned}I(Y_{uij}; \mathbf{Z}_{ij} \mid \mathcal{P}_1) - I(Y_{uij}; \mathbf{Z}_{ij,\perp} \mid \mathcal{P}_1) \\ = I(Y_{uij}; \mathbf{Z}_{ij,\parallel} \mid \mathbf{Z}_{ij,\perp}, \mathcal{P}_1).\end{aligned} \tag{25}$$

Proof. The eigenvalues of each soft projector are $1 - \lambda$ in the selected subspace and 1 in its orthogonal complement. Therefore, $\mathbf{P}_v$, $\mathbf{P}_t$, and $\mathcal{P}_\lambda$ are invertible whenever $\lambda < 1$. We consequently have deterministic mappings in both directions,

$$\widetilde{\mathbf{Z}}_{ij} = \mathcal{P}_\lambda\mathbf{Z}_{ij}, \qquad \mathbf{Z}_{ij} = \mathcal{P}_\lambda^{-1}\widetilde{\mathbf{Z}}_{ij}. \tag{26}$$

Applying the data-processing inequality in both directions proves Eq. (24). When $\lambda = 1$, the projected representation retains only $\mathbf{Z}_{ij,\perp}$. The mutual information chain rule gives

$$\begin{aligned}I(Y_{uij}; \mathbf{Z}_{ij} \mid \mathcal{P}_1) = I(Y_{uij}; \mathbf{Z}_{ij,\perp} \mid \mathcal{P}_1) \\ + I(Y_{uij}; \mathbf{Z}_{ij,\parallel} \mid \mathbf{Z}_{ij,\perp}, \mathcal{P}_1),\end{aligned} \tag{27}$$

which proves Eq. (25). In particular, a hard projection loses at most ϵ preference information when $I(Y_{uij}; \mathbf{Z}_{ij,\parallel} \mid \mathbf{Z}_{ij,\perp}, \mathcal{P}_1) \le \epsilon$.

Proposition 5 (Relative Exposure of Modality-Specific Evidence). Under the decomposition in Proposition 1, suppose that the repeated component $\mathbf{s}$ lies in the selected paired subspace, while $\mathbf{a}$ and $\mathbf{b}$ lie in its orthogonal complement. For $0 \le \lambda < 1$, the projected weighted fusion is

$$\mathbf{h}_\lambda = (1-\lambda)(\alpha+\beta)\mathbf{s} + \alpha\mathbf{a} + \beta\mathbf{b}. \tag{28}$$

Consequently, the ratio of modality-specific energy to repeated shared energy satisfies

$$\begin{aligned}\mathcal{R}_{\text{spec}}(\lambda) &= \frac{\alpha^2\mathbb{E}\|\mathbf{a}\|_2^2 + \beta^2\mathbb{E}\|\mathbf{b}\|_2^2}{(1-\lambda)^2(\alpha+\beta)^2\mathbb{E}\|\mathbf{s}\|_2^2} \\ &= \frac{\mathcal{R}_{\text{spec}}(0)}{(1-\lambda)^2}.\end{aligned} \tag{29}$$

Proof. Proposition 2 scales the selected component by $1-\lambda$ and leaves both orthogonal components unchanged. Substitution into the weighted fusion gives Eq. (28). Mutual uncorrelatedness removes the cross-terms in the expected squared norm, yielding Eq. (29). Hence CLEAR does not create new modality-specific information; it increases the relative exposure of the existing evidence that was previously overshadowed by repeatedly encoded shared variation.

Proposition 6 (Conditional Benefit for Pairwise Recommendation). Consider the dot-product scoring head commonly used with BPR. For a user–item triplet (u, i, j), decompose the unprojected pairwise score margin as

$$m_{uij}^{(0)} = m_{uij}^{\text{spec}} + m_{uij}^{\text{rep}}, \tag{30}$$

where m_{uij}^{spec} is induced by the preserved modality-specific representation and m_{uij}^{rep} is induced by the selected repeatedly encoded component. CLEAR gives

$$m_{uij}^{(\lambda)} = m_{uij}^{\text{spec}} + (1-\lambda)m_{uij}^{\text{rep}}. \tag{31}$$

If

$$\mathbb{E}\big[m_{uij}^{\text{rep}} \mid u, m_{uij}^{\text{spec}}\big] = 0, \tag{32}$$

then, for $\ell_{\text{BPR}}(m) = \log(1+\exp(-m))$,

$$\mathbb{E}\Big[\ell_{\text{BPR}}(m_{uij}^{(\lambda)})\Big] \le \mathbb{E}\Big[\ell_{\text{BPR}}(m_{uij}^{(0)})\Big]. \tag{33}$$

Without Eq. (32), the task-loss perturbation remains bounded by

$$\Big|\mathbb{E}\ell_{\text{BPR}}(m_{uij}^{(\lambda)}) - \mathbb{E}\ell_{\text{BPR}}(m_{uij}^{(0)})\Big| \le \lambda\mathbb{E}|m_{uij}^{\text{rep}}|. \tag{34}$$

Proof. Let $q = 1-\lambda$. For fixed u and m_{uij}^{spec},

$$\begin{aligned}m_{uij}^{\text{spec}} + qm_{uij}^{\text{rep}} &= q\left(m_{uij}^{\text{spec}} + m_{uij}^{\text{rep}}\right) \\ &\quad + (1-q)m_{uij}^{\text{spec}}.\end{aligned} \tag{35}$$

Because ℓ_{BPR} is convex, Jensen's inequality gives

$$\begin{aligned}&\mathbb{E}\big[\ell_{\text{BPR}}\big(m_{uij}^{\text{spec}} + qm_{uij}^{\text{rep}}\big) \,\big|\, u, m_{uij}^{\text{spec}}\big] \\ &\le q\,\mathbb{E}\big[\ell_{\text{BPR}}\big(m_{uij}^{\text{spec}} + m_{uij}^{\text{rep}}\big) \,\big|\, u, m_{uij}^{\text{spec}}\big] \\ &\quad + (1-q)\,\ell_{\text{BPR}}\big(m_{uij}^{\text{spec}}\big).\end{aligned} \tag{36}$$

Under Eq. (32), another application of Jensen's inequality shows that the final term is no greater than the conditional expected unprojected loss. Taking the outer expectation proves Eq. (33). Finally, $|\ell'_{\text{BPR}}(m)| \le 1$ and $|m_{uij}^{(\lambda)} - m_{uij}^{(0)}| = \lambda|m_{uij}^{\text{rep}}|$, which proves Eq. (34).

Interpretation. Equation (32) does not assert that every leading singular direction is irrelevant. It formalizes a sufficient case in which repeatedly encoded shared variation adds no incremental user-specific ranking evidence after the preserved representation is known. When this condition is only approximate, the invertibility result and the perturbation bound explain why soft attenuation is safer than hard removal, while validation selection of k and λ controls the empirical trade-off.

Representation Analysis Metrics

Sliced Wasserstein Distance

Sliced Wasserstein Distance (SWD) measures distributional separation through random one-dimensional projections. Given two sets of representations $\mathbf{V} = \{\mathbf{v}_i\}_{i=1}^N$ and $\mathbf{T} = \{\mathbf{t}_i\}_{i=1}^N$ mapped to the same d-dimensional analysis space, we compute

$$\begin{aligned}\text{SWD}(\mathbf{V}, \mathbf{T}) &= \frac{1}{L}\sum_{\ell=1}^{L} W_1(\mathbf{V}\boldsymbol{\theta}_\ell, \mathbf{T}\boldsymbol{\theta}_\ell), \\ &\boldsymbol{\theta}_\ell \in \mathbb{S}^{d-1}.\end{aligned} \tag{37}$$

where $\{\boldsymbol{\theta}_\ell\}_{\ell=1}^L$ are random unit vectors sampled uniformly from the unit sphere and W_1 is the one-dimensional Wasserstein distance computed from sorted samples. Larger SWD indicates stronger distributional separation between the two representation sets. We use it only as a representation-level diagnostic; it does not by itself imply higher task-relevant information.

Centered Kernel Alignment

We use linear Centered Kernel Alignment (CKA) (Kornblith et al. 2019) to quantify the structural similarity between pretrained visual and textual representations. For centered feature matrices $\bar{\mathbf{V}} = \mathbf{H}\mathbf{V}$ and $\bar{\mathbf{T}} = \mathbf{H}\mathbf{T}$, where $\mathbf{H} = \mathbf{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top$, linear CKA is

$$\text{CKA}(\mathbf{V}, \mathbf{T}) = \frac{\|\bar{\mathbf{V}}^\top\bar{\mathbf{T}}\|_F^2}{\|\bar{\mathbf{V}}^\top\bar{\mathbf{V}}\|_F\|\bar{\mathbf{T}}^\top\bar{\mathbf{T}}\|_F}. \tag{38}$$

CKA compares the item-wise relational structures encoded by the two modalities and is invariant to isotropic scaling and orthogonal transformations. A larger value indicates stronger cross-modal representational overlap. We therefore use CKA as a proxy for the degree of repeatedly encoded cross-modal structure, rather than as direct evidence that the shared information is task-irrelevant. In particular, CKA does not distinguish beneficial shared semantics from shared variation that may receive disproportionate emphasis during fusion; it only quantifies the amount of aligned relational structure available to be repeatedly encoded.

Table 5 reports the CKA values and the gains obtained by adding CLEAR to the same host backbone. As CKA decreases from Baby to Sports and MicroLens, the average relative gain across the four metrics also decreases from 3.28% to 2.40% and 1.47%, respectively. This monotonic trend is consistent with the mechanism that CLEAR tends to provide larger benefits when pretrained modalities exhibit

stronger representational overlap. Because this analysis contains only three datasets, we treat it as mechanism-consistent evidence rather than a statistical or causal conclusion.

CCA- and PLS-Based Direction Estimation

We additionally compare CLEAR with Canonical Correlation Analysis (CCA) and Partial Least Squares (PLS), which provide alternative criteria for estimating paired cross-modal directions. To ensure a controlled comparison, all variants use the same top-k rank and attenuation strength λ; only the direction-estimation procedure is changed.

Canonical Correlation Analysis. CCA identifies paired directions whose projected modality scores have maximum normalized correlation. Let

$$\begin{aligned}\mathbf{C}_{vv} &= \frac{1}{N-1}\bar{\mathbf{V}}^\top\bar{\mathbf{V}} + \epsilon\mathbf{I}_d,\\ \mathbf{C}_{tt} &= \frac{1}{N-1}\bar{\mathbf{T}}^\top\bar{\mathbf{T}} + \epsilon\mathbf{I}_d,\\ \mathbf{C}_{vt} &= \frac{1}{N-1}\bar{\mathbf{V}}^\top\bar{\mathbf{T}},\end{aligned} \tag{39}$$

where $\epsilon > 0$ is a regularization coefficient. CCA solves

$$\max_{\mathbf{a},\mathbf{b}} \frac{\mathbf{a}^\top\mathbf{C}_{vt}\mathbf{b}}{\sqrt{\mathbf{a}^\top\mathbf{C}_{vv}\mathbf{a}}\sqrt{\mathbf{b}^\top\mathbf{C}_{tt}\mathbf{b}}}. \tag{40}$$

Equivalently, the paired canonical directions are obtained from

$$\mathbf{C}_{vv}^{-\frac{1}{2}}\mathbf{C}_{vt}\mathbf{C}_{tt}^{-\frac{1}{2}} = \mathbf{U}_c\mathbf{\Gamma}\mathbf{W}_c^\top, \tag{41}$$

where $\mathbf{\Gamma}$ contains the canonical correlations. The corresponding directions are $\mathbf{A}_c = \mathbf{C}_{vv}^{-1/2}\mathbf{U}_c$ and $\mathbf{B}_c = \mathbf{C}_{tt}^{-1/2}\mathbf{W}_c$. Unlike CLEAR, which emphasizes directions with large cross-modal covariance energy, CCA normalizes each modality by its within-modality covariance and therefore emphasizes correlation independently of feature variance.

Partial Least Squares. PLS estimates paired latent directions by maximizing the covariance between the projected modality scores:

$$(\mathbf{a}_p, \mathbf{b}_p) = \arg\max_{\|\mathbf{a}\|_2=\|\mathbf{b}\|_2=1} \left(\mathbf{a}^\top\mathbf{C}_{vt}\mathbf{b}\right)^2. \tag{42}$$

In a sequential PLS procedure, subsequent direction pairs are extracted after deflating the modality representations with respect to the previously obtained latent scores. PLS therefore constructs cross-modal latent factors according to predictive covariance, whereas CCA optimizes normalized correlation.

For either alternative $m \in \{\mathrm{CCA}, \mathrm{PLS}\}$, we orthonormalize the selected visual and textual directions to obtain $\mathbf{Q}_v^{(m)}$ and $\mathbf{Q}_t^{(m)}$, and apply the same soft projections as CLEAR:

$$\begin{aligned}\mathbf{P}_v^{(m)} &= \mathbf{I}_d - \lambda\mathbf{Q}_v^{(m)}\mathbf{Q}_v^{(m)\top},\\ \mathbf{P}_t^{(m)} &= \mathbf{I}_d - \lambda\mathbf{Q}_t^{(m)}\mathbf{Q}_t^{(m)\top}.\end{aligned} \tag{43}$$

This design isolates the effect of direction estimation while keeping the projection rank, attenuation strength, recommendation backbone, and training objective unchanged.

Algorithm 1: Training Procedure with CLEAR

Input: Interactions $\mathbf{R}$; raw features $\mathbf{V}^{\mathrm{raw}}, \mathbf{T}^{\mathrm{raw}}$; host model g_Θ; original objective $\mathcal{L}_{\mathrm{base}}$; k, λ, and update interval τ

Output: Optimized model parameters Θ

for *epoch* $e = 1$ **to** *max_epochs* **do**
  if $e = 1$ ***or*** $e \bmod \tau = 0$ **then**
    $\mathbf{V} \leftarrow \phi_v(\mathbf{V}^{\mathrm{raw}})$, $\mathbf{T} \leftarrow \phi_t(\mathbf{T}^{\mathrm{raw}})$;
    $\boldsymbol{\mu}_v \leftarrow \frac{1}{N}\mathbf{V}^\top\mathbf{1}_N$, $\boldsymbol{\mu}_t \leftarrow \frac{1}{N}\mathbf{T}^\top\mathbf{1}_N$;
    $\bar{\mathbf{V}} \leftarrow \mathbf{V} - \mathbf{1}_N\boldsymbol{\mu}_v^\top$, $\bar{\mathbf{T}} \leftarrow \mathbf{T} - \mathbf{1}_N\boldsymbol{\mu}_t^\top$;
    $\mathbf{C} \leftarrow \mathrm{detach}\left(\frac{1}{N}\bar{\mathbf{V}}^\top\bar{\mathbf{T}}\right)$;
    $\mathbf{U}, \mathbf{\Sigma}, \mathbf{W}^\top \leftarrow \mathrm{SVD}(\mathbf{C})$;
    Cache $\boldsymbol{\mu}_v$, $\boldsymbol{\mu}_t$, $\mathbf{U}_k$, and $\mathbf{W}_k$;
  end
  foreach *interaction mini-batch* $\mathcal{B}$ **do**
    Encode the item features required by g_Θ and center them using the cached global means;
    $\widetilde{\mathbf{V}} \leftarrow \bar{\mathbf{V}} - \lambda(\bar{\mathbf{V}}\mathbf{U}_k)\mathbf{U}_k^\top$;
    $\widetilde{\mathbf{T}} \leftarrow \bar{\mathbf{T}} - \lambda(\bar{\mathbf{T}}\mathbf{W}_k)\mathbf{W}_k^\top$;
    Compute $\mathcal{L}_{\mathrm{base}}$ on $\mathcal{B}$ with $g_\Theta(\mathbf{R}, \widetilde{\mathbf{V}}, \widetilde{\mathbf{T}})$;
    $\Theta \leftarrow \Theta - \eta\nabla_\Theta\mathcal{L}_{\mathrm{base}}$;
  end
end

Implementation and Complexity

Backbone Integration and Objective

CLEAR is applied after the modality encoders and before the original recommendation backbone. For each backbone, the projected features $\widetilde{\mathbf{V}}$ and $\widetilde{\mathbf{T}}$ replace the unprojected modality features at the same input interface. All backbone-specific propagation, fusion, scoring, and auxiliary components remain unchanged. Each backbone therefore retains its original training objective.

For backbones optimized with Bayesian Personalized Ranking (BPR) (Rendle et al. 2012), the recommendation loss is

$$\mathcal{L}_{\mathrm{BPR}} = -\sum_{(u,i,j)\in\mathcal{O}} \log\sigma(\hat{y}_{ui} - \hat{y}_{uj}), \tag{44}$$

where $\mathcal{O} = \{(u,i,j) : r_{ui} = 1,\ r_{uj} = 0\}$. The complete objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{BPR}} + \gamma\|\Theta\|_2^2. \tag{45}$$

For backbones with self-supervised or auxiliary objectives, we retain their complete original losses. CLEAR introduces no additional loss term.

Training Procedure

The global feature means and paired singular directions are updated every τ epochs and cached between updates. The covariance used to construct the projection is detached from the computation graph; during ordinary training, gradients still flow through the encoded and projected features.

Table 5: Relationship between cross-modal representational similarity and the gains from CLEAR. CKA is computed from pretrained visual and textual features. Base denotes the same host backbone and training setting with the CLEAR projection disabled.

Dataset	CKA	Base				CLEAR				Gain (%)			
		R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20
Baby	0.26	0.0659	0.1017	0.0352	0.0444	0.0687	0.1045	0.0364	0.0456	4.25	2.75	3.41	2.70
Sports	0.20	0.0740	0.1120	0.0401	0.0498	0.0756	0.1146	0.0411	0.0511	2.16	2.32	2.49	2.61
MicroLens	0.12	0.0795	0.1181	0.0412	0.0510	0.0807	0.1194	0.0419	0.0518	1.51	1.10	1.70	1.57

Time and Space Complexity

Constructing the centered cross-modal covariance requires $\mathcal{O}(Nd_vd_t)$ time. A full SVD requires $\mathcal{O}(d_vd_t\min(d_v,d_t))$ time, while a truncated top-k SVD can reduce this cost to approximately $\mathcal{O}(d_vd_tk)$. Using

$$\widetilde{\mathbf{V}} = \bar{\mathbf{V}} - \lambda(\bar{\mathbf{V}}\mathbf{U}_k)\mathbf{U}_k^\top \tag{46}$$

avoids constructing a dense $d_v \times d_v$ projection matrix and requires $\mathcal{O}(Nd_vk)$ time. The textual projection analogously requires $\mathcal{O}(Nd_tk)$. The additional memory cost is $\mathcal{O}(d_vd_t + k(d_v+d_t))$ for the covariance matrix and paired bases. These quantities are updated only once every τ epochs and cached across mini-batches. When $d_v = d_t = d$, the dense upper bounds reduce to $\mathcal{O}(Nd^2 + d^3)$ time and $\mathcal{O}(d^2)$ additional space, while the low-rank application above reduces the per-update projection cost to $\mathcal{O}(Ndk)$.

Additional Generalization Results

To complement the generalization analysis on Baby in Table 4, Tables 7 and 8 report the corresponding results on Sports and Clothing, respectively. We use the same evaluation protocol and report Recall@10 (R@10) and NDCG@10 (N@10). Base denotes the original backbone, while + CLEAR denotes the same backbone equipped with the proposed projection module.

Details of Experimental Setup

Datasets

- **Baby**: Amazon Baby contains user interactions with products such as strollers, toys, and childcare items.
- **Sports**: Amazon Sports contains interactions with sports and outdoor products, including equipment, apparel, and accessories.
- **Clothing**: Amazon Clothing contains interactions with clothing, shoes, and jewelry products.
- **MicroLens**: MicroLens is a micro-video recommendation benchmark. The processed subset used in our experiments contains 98,129 users, 17,228 items, and 705,174 interactions.

We apply 5-core filtering to the three Amazon datasets. Their visual features are the commonly used 4,096-dimensional pre-extracted features, and their textual features are encoded into 384-dimensional vectors with a pretrained sentence transformer. For MicroLens, we follow the public benchmark protocol to construct the visual and textual item features.

Evaluation Protocol and Implementation

We randomly split interactions into training, validation, and test sets with an 8:1:1 ratio and evaluate Recall@K and NDCG@K for $K \in \{10, 20\}$. Hyperparameters are selected according to validation performance. For CLEAR, we search $\lambda \in \{0.3, 0.5, 0.7, 0.8, 0.9\}$ and $k \in \{1, 2, 4, 8, 16, 20\}$. Experiments are implemented with PyTorch and conducted on a single NVIDIA RTX 4090 GPU with 24 GB memory.

Baselines

We compare against 14 representative baselines in two groups.

General collaborative filtering methods.

- **MF-BPR** (Rendle et al. 2012): Matrix factorization optimized with Bayesian Personalized Ranking for implicit feedback.
- **LightGCN** (He et al. 2020): A simplified graph convolutional recommender that removes feature transformations and nonlinear activations from neighborhood propagation.
- **LayerGCN** (Zhou et al. 2023a): A layer-refined GCN that alleviates over-smoothing by adaptively combining representations across propagation layers.

Multimodal recommendation methods.

- **VBPR** (He and McAuley 2016): A visual recommendation model that incorporates item visual features into Bayesian Personalized Ranking.
- **MMGCN** (Wei et al. 2019): A multimodal graph convolutional network that constructs modality-specific user–item graphs.
- **DualGNN** (Wang et al. 2021a): A dual graph model that combines user–item interactions with user co-occurrence structures.
- **LATTICE** (Zhang et al. 2021): A graph-based method that learns latent item–item structures from multimodal similarities.
- **SLMRec** (Tao et al. 2022): A self-supervised framework using feature dropout and masking augmentations for multimodal recommendation.

Table 6: Memory consumption and training time per epoch for the compared models under the same experimental environment.

Dataset	Metric	General CF models			Multimodal models			
		MF-BPR	LightGCN	VBPR	MMGCN	LATTICE	FREEDOM	CLEAR
Baby	Memory (GB)	1.59	1.69	1.89	2.69	4.53	2.13	3.11
	Time (s/epoch)	0.86	1.81	1.04	6.35	3.42	2.58	2.30
Sports	Memory (GB)	2.00	2.24	2.71	3.91	19.93	3.34	4.12
	Time (s/epoch)	1.24	3.74	1.67	20.88	13.59	4.52	4.77
Clothing	Memory (GB)	2.16	2.43	3.02	4.24	28.22	4.15	4.43
	Time (s/epoch)	1.52	4.58	2.02	26.13	23.96	5.74	5.34

Table 7: Generalization analysis on the Sports dataset.

	Base		+ CLEAR		Gain (%)	
Backbone	**R@10**	**N@10**	**R@10**	**N@10**	**R@10**	**N@10**
VBPR	0.0558	0.0307	**0.0614**	**0.0338**	10.04	10.10
FREEDOM	0.0717	0.0385	**0.0734**	**0.0394**	2.37	2.34
MENTOR	0.0736	0.0399	**0.0756**	**0.0411**	2.72	3.01

Table 8: Generalization analysis on the Clothing dataset.

	Base		+ CLEAR		Gain (%)	
Backbone	**R@10**	**N@10**	**R@10**	**N@10**	**R@10**	**N@10**
VBPR	0.0281	0.0158	**0.0315**	**0.0174**	12.10	10.13
FREEDOM	0.0629	0.0341	**0.0644**	**0.0350**	2.38	2.64
MENTOR	0.0659	0.0357	**0.0675**	**0.0358**	2.43	0.28

- **MICRO** (Zhang et al. 2022): A multimedia recommendation method that mines modality-aware latent item–item structures and employs contrastive learning for fine-grained multimodal fusion.
- **BM3** (Zhou et al. 2023b): A bootstrap self-supervised method based on dropout perturbations of multimodal representations.
- **FREEDOM** (Zhou and Shen 2023): A method that freezes item–item graph structures and applies degree-sensitive edge pruning for denoising.
- **MGCN** (Yu et al. 2023): A multi-view graph model that purifies modality features with behavioral information and performs behavior-aware fusion.
- **LGMRec** (Guo et al. 2024): A method that captures local and global multimodal graph structures for recommendation.
- **MENTOR** (Xu et al. 2025): A multi-level self-supervised framework that improves cross-modal alignment under sparse interaction supervision.